

Fight detection with spatial and channel wise attention-based ConvLSTM model

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Abstract

An automated detection of aggressive and violent behaviour in videos has immense potential. It enables efficient online content filtering by restricting access to extreme content and also, when integrated with security systems, helps to monitor violence in surveillance videos. In this work, a convolutional neural network is combined with the proposed Spatial and Channel wise Attention-based ConvLSTM encoder (SCan-ConvLSTM). The proposed architecture performs an efficient spatiotemporal fusion of the features extracted from the video sequences containing fight scenes. In order to focus selectively on regions of utmost importance, this blended attention mechanism adjusts the weights of outputs in different locations and across different channels. This recurrent attention mechanism enhances the sequential refinement of activation maps and boosts the model performance. Finally, the experimental results have been presented that show the proposed architecture achieves superior results on the benchmark datasets (RWF-2000, Violent-flow, Hockey-fights, and Movies).

KEYWORDS

attention, convolutional neural networks, deep learning, fight detection, LSTM, violence

1 | INTRODUCTION

Security agencies and defence organizations spend a large amount of their budget to deploy personnel for manual inspection of the surveillance videos. However, it is practically not infeasible to monitor round the clock, thus making the whole system inefficient. Automation of surveillance systems can minimize response time by alerting administrations and preventing acts of mass destruction. Apart from the installation of such systems in public places and prisons, it can function efficiently in filtering violent videos from websites containing fight scenes that are often responsible for spreading enmity among people. In this work, fight between individuals is distinguished from the other actions by learning the mechanics around the movement of body parts. Usually, in fight scenes, people move in random directions at an unusual speed, that is, the motion dynamics of a person or a crowd involved in violence have a higher degree of variability over time (Chang et al., 2020).

Visual perception is one of the most important aspects of the brain system. It enables the individual to focus on relevant regions in the environment to extract crucial information. This information becomes more important when we are dealing with violence in crowded or non-crowded scenes. In such cases, there is a need to attend selectively on specific parts of the field of view. The attention mechanism discards irrelevant features, while focusing on a small but relevant part of the frame. It helps to build a better representation of the scene of a fight. Recently, attention mechanism has been used to solve vision problems such as activity recognition (Sharma et al., 2015), image (Yan et al., 2021) and video captioning (Yan et al., 2020), and machine translation (Vaswani et al., 2017). Sequential data with complex dynamics and appearance variations, demands efficient spatio-temporal analysis. The fight detection problem has been addressed by defining ConvLSTM (He et al., 2015) model. It preserved the 2D structure of the input feature maps, while performing temporal analysis. However, system inability to retrieve information by focusing area of interest with key features, is a big challenge (Hanson et al., 2019; Sudhakaran & Lanz, 2017). Sharma et al. (2015) presented a soft attention model to classify different activities by weighing importance to relevant pieces of evidences. Such systems (Sharma et al., 2015) are helpful for

healthcare services and autonomous vehicles (Kim & Canny, 2017). Hu et al. (2017) defined a squeeze-and-excitation (SE) block that used channel-wise statistics of feature maps to suppress irrelevant information between different channels. However, the squeeze-and-excitation (SE) block ignored the significance of spatial attention. Inspired by the work of Chen et al. (2016), we have proposed spatial and channel-wise attention based ConvLSTM (SCan-ConvLSTM) framework for violence detection in crowd. It integrates the attention mechanisms with the recurrent learning mechanism named as ConvLSTM model. Where, spatial attention responds to 'where to attend?' and channel-wise attention responds to 'what to attend?'. The main contributions of the work are as follows:

- A novel spatial attention and channel-wise attention-based ConvLSTM (SCan-ConvLSTM) framework is proposed to detect fight scenes in crowd.
- The proposed framework enhances the recurrent mechanism and improve the model's ability to classify fight scenes in videos.
- To exhibit the efficacy of the proposed framework, extensive experiments are performed on four publically available benchmark datasets, namely RWF-2000, Violent-flow, Hockey-fights, and Movies.

The article is organized as follows. Section 2 reviews the related work in the field of fight detection. Further, in Section 3, we present our proposed spatio-temporal architecture based on the novel SCan-ConvLSTM. Section 4 describes the four datasets used, along with the discussion of the results using different variants of ConvLSTM and the comparison with the state-of-the-art methods. In Section 5, a conclusion is drawn about the proposed work, and future work is discussed.

2 | RELATED WORK

The issue of detecting violence from audio, visual, or text data has been addressed in the past. Activities of violence such as cyber victimization, dating aggression, internet banging that uses offensive language could be detected with sentiment analysis using natural language processing (Coletto et al., 2018). Moreover, fusion models have been proposed to detect violence in movie scenes using visual and audio feature representations (Peixoto et al., 2020). However, these approaches have limited applicability to surveillance videos that lack audio information.

Traditional approaches for fight detection use hand-crafted features (Bilinski & Bremond, 2016; Gracia et al., 2015; Hassner et al., 2012). These approaches are either focused on bag-of-words or local features. Gracia et al. (2015) relied on motion blob features for the recognition of a fight from a video sequence. It used distance between the blobs, area, and perimeter to categorize blobs. Violent Flows (ViF) descriptor by Hassner et al. (2012) encrypted apparent relative motion among the adjacent frames as optical flow to detect violence. Further, Mohammadi et al. (2015) introduced a new descriptor based on substantial derivatives. Such approaches ignored spatio-temporal information of the features. To overcome this, Bilinski and Bremond (2016) proposed an extension of Improved Fisher Vectors (IFV) that used the spatiotemporal points and local features. Summed area table data structure helped to speed up the violence detection in videos.

Recently, several CNN-based approaches (Ding et al., 2014; Hanson et al., 2019; Irfanullah et al., 2022; Serrano et al., 2018; Sudhakaran & Lanz, 2017; Ullah et al., 2019) have been proposed for fight detection. In contrast to deep learning-only approaches, Serrano et al. (2018) defined a hybrid fight detection framework that integrated Hough forests based handcrafted features and 2D CNNs based deep features. In order to encode temporal information, some of the recent works (Cheng et al., 2019; Irfanullah et al., 2022; Li et al., 2019; Ullah et al., 2019) have used 3DConvNets. The work (Ullah et al., 2019) selectively chose frames containing persons and pass them to the 3DCNN using pre-trained MobileNet. In this process, the short temporal size of the input significantly affects the performance of 3DCNNs, which leads to the inability to detect violent actions at the full temporal extent. In addition, 3DCNNs have huge computational costs and require large amounts of data for training. Sudhakaran and Lanz (2017) used 2DCNN with ConvLSTM that enabled the spatiotemporal encoding within the cell. This work was further extended by the application of Bidirectional-ConvLSTM (Hanson et al., 2019), to encode spatio-temporal features effectively. However, in the case of crowded scenes, previous approaches are unable to make a distinction between spectators and those involved in the fight. The aforementioned approaches pay less attention to improve spatial and channel wise information, this may lead to learning irrelevant information for training models, and ignoring salient information. Irfanullah et al. (2022) utilized MobileNet model for intelligent reconnaissance video examination in real time to cater a practical solution for the difficult human undertaking. A recent work (Amin et al., 2023), attempted to define Quantum CNNs to address the sufficiently large data or model dimension limitation. It classified the number of violent robberies such as armed thefts containing handguns or knives, and robberies displaying varying levels of viciousness. Another work (Khaire & Kumar, 2022) developed a multi-modal semi-supervised deep learning based CNN-BiLSTM autoencoder framework for video anomaly detection using RGB-D videos in critical surveillance environments like Bank-ATMs.

Now, the works (Le & Kim, 2023; Li et al., 2022) are performing attention based anomaly detection. The work (Li et al., 2022) defined a future frame prediction framework and a Multiple Instance Learning (MIL) framework by leveraging attention schemes to learn anomalies, that is, such as intrusion, crowd, accident, weapon, arson using multi-view videos. The work has been experimented with only 170 multi-view videos. Whereas Le and Kim (2023) defined deep convolutional neural network-based encoder and a multi-stage channel attention-based decoder in unsupervised

learning environment. In this direction, Ullah et al. (2023) developed a sequential attention mechanism for weakly supervised video anomaly detection.

The proposed work presents a Spatial and Channel-wise Attention based framework (SCan-ConvLSTM) to detect fights in videos using ConvLSTM. Where, spatial attention emphasizes specific locations to put attention at and highlight the frames, and channel-wise attention mechanism pays attention to discriminative features of a frame to obtain contextual information.

3 | PROPOSED METHODOLOGY

In this section, the proposed Spatio-temporal Scan-ConvLSTM fight detection network is discussed, as shown in Figure 1. First, spatial features are extracted using the state-of-the-art ResNet-50 (He et al., 2015) architecture. Spatial and temporal attention mechanism is applied on extracted features by defining a variant of ConvLSTM learning model, that is, Scan-ConvLSTM.

3.1 | Base model

Long Short Term Memory (LSTMs) (Hochreiter & Schmidhuber, 1997) sequential learning model has been defined to solve the problem of vanishing gradients faced by Vanilla RNNs. LSTMs store the information in a state memory cell. However, it is not suitable for sequential spatial information as there might be a loss of correlation between the spatial and temporal information. CNNs (Chang et al., 2020) have been popularly used to exploit the spatial information. Different variants of CNNs, that is, 3DCNN (Sudhakaran & Lanz, 2017), C3D (Hanson et al., 2019), Distributed CNN (DCNN) (Hanson et al., 2019) are presented to encrypt spatio-temporal features in videos. However, it is observed that explicit LSTM model encodes temporal information in complex scenes better than 3D-CNN architecture. Chang et al. (2020) amalgated both the concepts and developed ConvLSTM architecture to learn spatiotemporal features from long-term sequences without changing the features to one-dimensional. In the proposed work the ConvLSTM is used as a baseline for experimentation.

The fully connected layers in conventional LSTM model are replaced with the convolutional layers, as shown in Figure 2. The key operations of the ConvLSTM are shown in Equations (1)–(6),

$$i_t = \text{sig}(\omega_{xi} * X_t + \omega_{hi} * H_{t-1} + b_i), \quad (1)$$

$$f_t = \text{sig}(\omega_{xf} * X_t + \omega_{hf} * H_{t-1} + b_f), \quad (2)$$

$$O_t = \text{sig}(\omega_{xo} * X_t + \omega_{ho} * H_{t-1} + b_o), \quad (3)$$

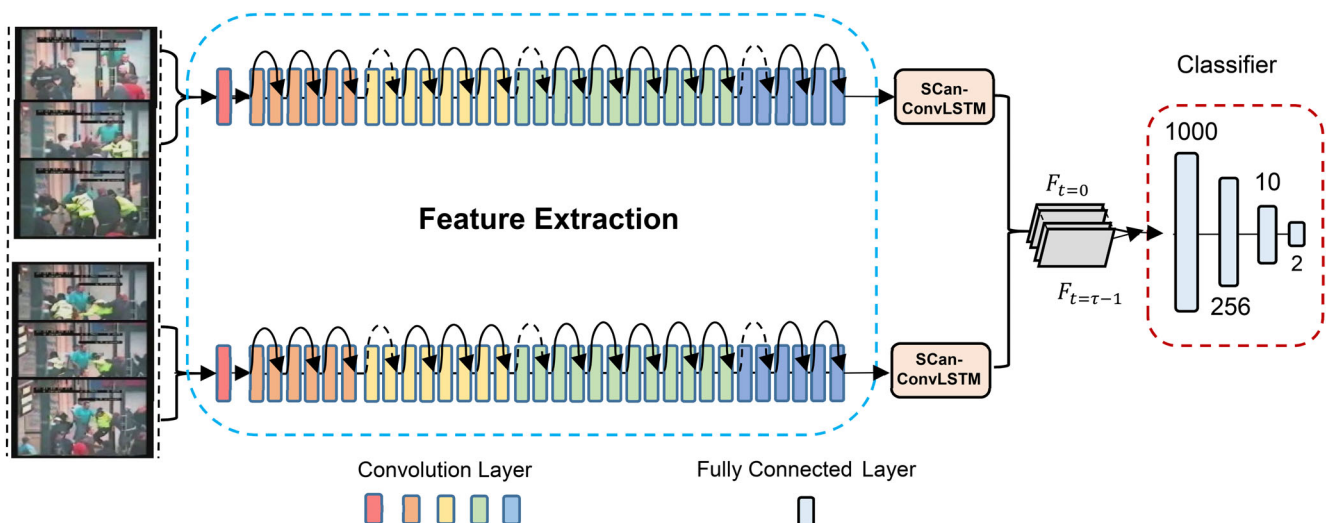


FIGURE 1 Illustration of proposed approach in three parts: ResNet50 architecture for spatial feature extraction with fixed input size of 224×224 , ConvLSTM variant to perform temporal encoding and the classifier to perform the detection of fight scenes in videos.

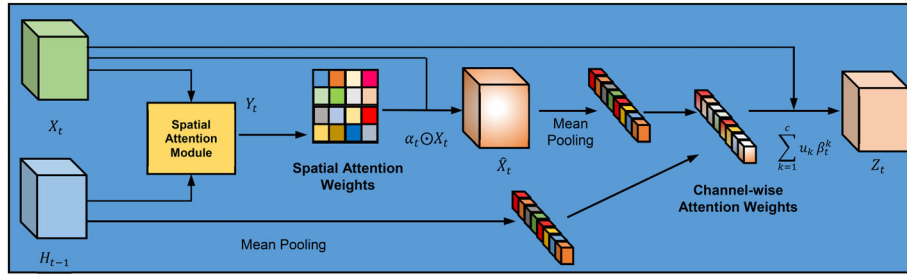


FIGURE 2 Illustration of proposed ConvLSTM variant, SCan-ConvLSTM. Here, spatial attention is applied, followed by channel-wise attention mechanism to the inputs of ConvLSTM model.

$$\dot{c}_t = \tanh(\omega_{xc} * X_t + \omega_{hc} * H_{t-1} + b_c), \quad (4)$$

$$\mathcal{C}_t = i_t \odot \dot{c}_t + \mathcal{F}_t \odot \mathcal{C}_{t-1}, \quad (5)$$

$$H_t = O_t \odot \tanh(\mathcal{C}_t), \quad (6)$$

where, sig is used for sigmoid function, * denotes the convolutional operator, Hadamard product is represented by $\odot$ symbol, $\omega_{xif,o}$ and $\omega_{hif,o}$ are 2D convolution kernels, $\dot{c}_t$ denotes candidate memory. X_t is the input at time t while $i_t, \mathcal{F}_t, O_t$ are gate functions and b_i, b_o, b_f are the biases of input, output and forget gates. H_t and $\mathcal{C}_t$ are hidden state and cell state, respectively, and these are updated at each time step.

3.2 | SAtt-ConvLSTM: Spatial attention based ConvLSTM

Spatial attention mechanism (Li et al., 2018) applied on the inputs of the ConvLSTM cell, focuses on a specific Region of Interest (ROI). Two dimensional convolution kernel is defined mathematically as given in Equation (7). A convolution operation converts input to state and state to state and captures the spatial information of attributes better than standard LSTM. Hence, provides better the correlation between ROI attributes

$$Y_t = \omega_Y * \tanh(\omega_{xa} * X_t + \omega_{ha} * H_{t-1} + b_A), \quad (7)$$

where, Y_t and X_t are three dimensional tensors; the first dimension represents temporal information, the second and third dimension represent the rows and columns of spatial information. $\tanh(\cdot)$ represents the hyperbolic tangent function. H_{t-1} is a hidden state. ω_Y, ω_{xa} and ω_{ha} are the weights and b_A is the bias. The next step is to generate a 2D attention map, given by Equation (8) that identifies spatial relationship between pixels of the feature map Y_t .

$$\alpha_t^{ij} = \frac{P(X_t, \dot{H}_{t-1} | \text{att}^{ij}) P(\text{att}^{ij})}{P(X_t, \dot{H}_{t-1})} = \frac{\exp(Y_t^{ij})}{\sum_i \sum_j \exp(Y_t^{ij})}, \quad (8)$$

where, α_t^{ij} is the element of α_t which is the 2D attention map generated by applying spatial attention mechanism. At last, elementwise sum operation is performed on attention map α_t and original input feature X_t to obtain the resultant feature map $\hat{X}_t$.

$$\hat{X}_t = \alpha_t \odot X_t. \quad (9)$$

3.3 | SCan-ConvLSTM: Spatial and channel wise attention based ConvLSTM

In the proposed work, we define a novel spatial and channel-wise attention based feature learning mechanism at the inputs of ConvLSTM (SCan-ConvLSTM) as illustrated in Figure 2. It applies spatial attention followed by the channel-wise attention. Spatial attention here pays attention to relevant spatial regions, while channel-wise attention helps suppress irrelevant information across different channel dimensions. Channel-wise attention enhances the correlation between the discriminative features among the different channels. Our spatial and channel-wise



attention block receives two inputs, input feature map, $\hat{X}_t$ as obtained in Equation (9) and the hidden state, H_{t-1} . Here $\hat{X}_t$ and H_{t-1} can be defined using Equations (10) and (11),

$$\hat{X}_t = [u_1, u_2, \dots, u_c], \quad (10)$$

$$H_{t-1} = [h_1, h_2, \dots, h_c], \quad (11)$$

where, u_i and h_i is a feature map of size $W \times H$ along the dimension of i th channel dimension and c are the numbers of channels. Channel feature $\hat{X}'_t$ and H'_{t-1} are obtained by applying mean pooling as given below:

$$\hat{X}'_t = [v_1, v_2, \dots, v_c], \quad (12)$$

$$H'_{t-1} = [g_1, g_2, \dots, g_c], \quad (13)$$

where, v_i and g_i are the scalar entities obtained by pooling operation over u_i and h_i . Now, the attention weights of the i th channel, β_t^i are calculated as in Equation (14),

$$\beta_t^i = \text{sig} \left(W_{ca2} * \delta \left(W_{ca1} * [\hat{X}'_{t,i}, H'_{t-1,i}] + b_{ca1} \right) + b_{ca2} \right), \quad (14)$$

where, $W_{ca2} \in \mathbb{R}^{\tilde{c} \times c \times 1 \times 1}$ and $W_{ca1} \in \mathbb{R}^{c \times \tilde{c} \times 1 \times 1}$ are the weights and $b_{ca1} \in \mathbb{R}^{\tilde{c}}$ and $b_{ca2} \in \mathbb{R}^c$ are the biases where $\tilde{c} < c$ and $\text{sig}(\cdot)$ and $\delta(\cdot)$ are the sigmoid and RELU function, respectively. Now, we calculate the final modified input feature map, Z_t as shown in Equation (15) using Hadamard product for the enhanced feature discriminability,

$$Z_t = \sum_{i=1}^c u_i \beta_t^i. \quad (15)$$

The rest expressions are similar to the Equations (1)–(6) of the ConvLSTM model.

4 | EXPERIMENTS

An experimental evaluation of the proposed methodology is carried out, in this section. The experiments are conducted on four benchmark datasets for fight detection, including RWF-2000 Dataset (Cheng et al., 2021), Hockey Fight Dataset (Bermejo Nieves et al., 2011), Crowd Violence Dataset (Hassner et al., 2012), and Movies Dataset (Bermejo Nieves et al., 2011) followed by detailed discussion of the results.

RWF-2000 Dataset: Cheng et al. (2019) proposed the RWF-2000 dataset that consist of 2000 fight and non-fight videos. The length of each video is 5 s, captured using surveillance cameras. Half of the videos include violent behaviours, while others belong to nonviolent activities. It is the most challenging and largest dataset available as it is not limited to any specific type of violence, including explosion, shooting, attack, and so forth.

Hockey Fight Dataset: It is a collection of 500 violent and 500 non-violent videos each filmed on an ice hockey rink during the National Hockey League. Each video is limited to approximately 50 frames with a resolution of 360×288 pixels. Fight between two players to a few players with a similar background has been shown in the video sequences, in Figure 3b.

Crowd Violence Dataset: It is a collection of 246 Violence videos in the crowd, collected from YouTube. It is a real-time database for violent crowd behaviour with a non-uniform background. Half of the videos include violent behaviours, and others belong to nonviolent activities. The duration of each video clip varies between 1.04 and 6.52 s. It provides wide range of challenging real world viewing uncontrolled, in-the-wild conditions with wide range of scene types, video qualities and surveillance scenarios.

Movies Dataset: The Dataset was built specifically for the classification of fight and non-fight sequences from movies. It consists of 200 videos clips in which fights were extracted from action movies and non-fight videos were extracted from public action recognition datasets. It is not a uniform dataset like RWF-2000, Hockey dataset and Crowd Violence dataset. Therefore, to keep it uniform and balanced for experimentation, equal number of fight and non-fight videos are chosen for framework learning.

From Table 1, it can be observed that, each dataset covers different types of violence encountered in day-to-day life, that is, violence in movies, surveillance, hockey game, and natural.

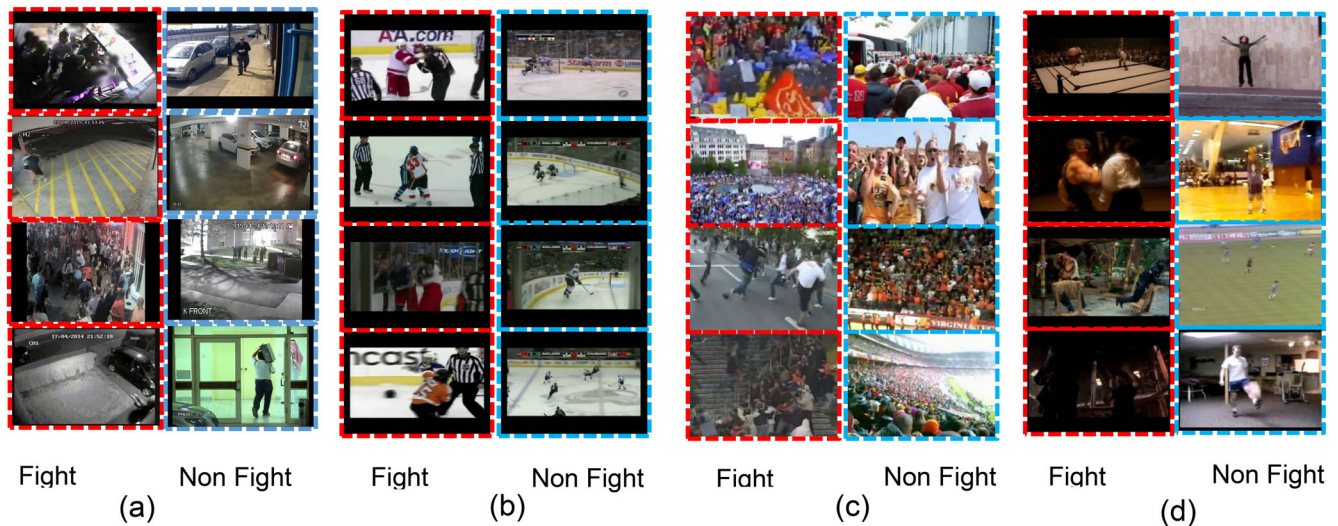


FIGURE 3 The illustration of some frames from the four benchmark datasets: (a) RWF-2000, (b) the Hockey Fight, (c) the Crowd Violence, and (d) the Movies.

TABLE 1 Datasets description.

Dataset	Data scale	Length/clip (s)	Resolution	Annotation	Scenario
Crowd Violence	246 clips	1.04–6.52	Variable	Video level	Natural
Movies Fight	200 clips	1.6–2	720 × 480	Video level	Movie
Hockey Fight	1000 clips	1.6–1.96	360 × 288	Video level	Hockey games
RWF-2000	2000 clips	5	Variable	Video level	Surveillance

Note: The 'Natural' represents that videos are from realistic scenes, but recorded by hybrid types of devices (e.g., mobile cameras, car-mounted cameras).

TABLE 2 Comparison of the different variants of ConvLSTM on benchmark datasets.

Dataset	Method					
	ConvLSTM		SAtt-ConvLSTM		SCan-ConvLSTM	
	Accuracy	F1	Accuracy	F1	Accuracy	F1
RWF-2000	79.12% ± 0.34%	0.796	81.37% ± 0.31%	0.831	84.32% ± 0.47%	0.84
Hockey-Fight	98.21% ± 0.46%	0.987	99.46% ± 0.41%	0.993	99.83% ± 0.52%	0.996
Crowd Violence	96.35% ± 0.48%	0.966	100% ± 0%	1.000	98.753 ± 0.63%	0.988
Movies	100% ± 0%	1.000	100% ± 0%	1.000	100% ± 0%	1.000
Gen_Test_Case	89.96% ± 0.49%	0.899	96.85% ± 0.21%	0.967	98.41 ± 0.57%	0.983

Note: Bold indicates highest accuracy by our model.

For the experimentation, 5-fold cross validation scheme is used. Sample images of a fight and non-fight categories of each dataset, are shown in Figure 3. The performance of the proposed violence detection model is measured in terms of Accuracy and F1 score. As per evaluation strategies used in the evaluation of previous literature work, accuracy is averaged over a radius of 10 epochs around the epoch where the model achieved the best result and the standard deviation of the accuracies is calculated. In the experimentation, RGB frames of video clips are resized to 224×224 dimension, followed by sampling and normalization to the range of 0 and 1. Each model is trained with the RMSprop optimizer and the initial learning rate is chosen at 0.0001 with a mini-batch size of eight clips.

To exhibit the generalisability of the proposed framework, experimentation is extended with a testing case: Gen_Test_Case, in which samples from all four datasets are merged. This kind of pool covers diverse set of violence and non-violence videos from sports ground, movies, crowded

background and quite close to real world video instances. This resulted in a total of 3446 video clips with 1723 fight clips and 1723 non-fight clips. The performance of the Gen_Test_Case is also measured in terms of accuracy and F1 score.

This study performs a detailed comparison of our proposed SCan-ConvLSTM with the following variants of LSTM: ConvLSTM, Spatial Attention-based ConvLSTM (SAtt-ConvLSTM) for all four datasets individually and combined as Gen_Test_case, as reported in Table 2. To be specific, we have used common settings for each ConvLSTM variant on the same dataset for a fair comparison.

From Table 2, it is clearly evident that SCan-ConvLSTM learning model exhibit significant overall improvement in classification accuracy (bold text) over the different variants. SAtt-ConvLSTM, incorporated only spatial attention mechanism at the inputs of ConvLSTM and performed slightly better than the original ConvLSTM. However, the performance of SAtt-ConvLSTM is lesser than the proposed SCan-ConvLSTM learning model. The key reason behind superior performance of SCan-ConvLSTM is that it identifies the relevant regions where there is a density of people and differentiates between the background (i.e., surroundings) and foreground (i.e., People) simultaneously. In addition, Gen_Test_case test case validates that the proposed work can handle the complexity of the diverse merged samples covering all variations of violent videos with superior performance.

An accuracy of 99.9% and 100% for Hockeyfight and Movie dataset respectively, is observed due to the lack of diversity of samples in each class of fight and non-fight videos. The Movies dataset contains fight sequences between a few individuals. Whereas, the Crowd Violence dataset contains relatively complex crowded, violent scenes.

For the RWF-2000 dataset, the accuracy achieved using SCan-ConvLSTM is better than the ConvLSTM and SAtt-ConvLSTM. However, the performance of the proposed model for Movie fight, Hockey Fight and Crowd Violence datasets, is superior than RWF-2000 dataset. This is attributed to the fact that the dataset consists of diversified samples of violence, including robbery, explosion, shooting, attack, and so forth that are not limited to just fighting between a crowd or a few people. ROC Curves are illustrated in Figure 4 using different ConvLSTM variants on the three benchmark datasets. We do not evaluate the curve for the Movie dataset as the model achieved 100% accuracy for each ConvLSTM variant. The potential of using both spatial and channel wise attention mechanism in ConvLSTM model can be demonstrated from the ROC curves.

A comparative analysis is outlined in Tables 3 and 4 by covering both handcrafted and deep features based violence detection in the videos. Our proposed deep learning model achieves state-of-the-results on Hockey Fight Dataset, Crowd Violence dataset, and the Movies Dataset with an accuracy of $99.16\% \pm 0.36\%$, $97.65\% \pm 0.82\%$, and $100\% \pm 0\%$, respectively.

From Table 4, it is observed that the proposed work is able to outperform ConvLSTM, C3D, I3D state-of-the-art for RWF-2000 dataset. Whereas, Flow Gated Network (Cheng et al., 2019) and SPIL (Su et al., 2020) are performing marginally higher. SPIL (Su et al., 2020) has utilized skeleton poses to understand the scene violence carefully. It generates 3D point clouds first and learns the joint interactions of people in the frame. SPIL mask has improved the performance by 2%–3% from SPIL-space (Su et al., 2020). Flow Gated Network (Cheng et al., 2019) used both RGB stream and optical flow maps to learn scene semantics for violence prediction. Though work (Cheng et al., 2019) outperforms other works by large margin, however, its two stream learning model is computationally expensive from practical application. On the otherside, the proposed work only utilized RGB stream to learn the violence events in the videos. It makes SCan-ConvLSTM model suitable for practical surveillance application for violence detection. In Figure 5, saliency prediction results are visualized using attention weights on the sample videos from RWF-2000 and Hockey Fight datasets.

Notably, the proposed SCan-ConvLSTM learning model with ResNet50 based features, is efficient for violence detection. It allows highly accurate and efficient detection of fighting scenes from a video sequence. To the best of our knowledge, this is the first study that uses both spatial attention and channel-wise attention mechanism for the task of fight detection in videos.

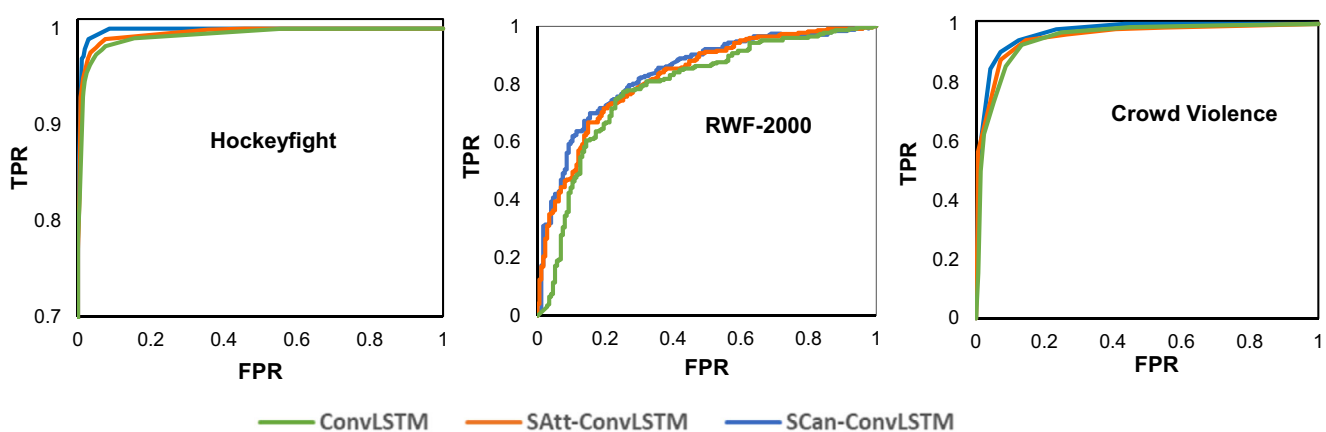


FIGURE 4 ROC curves depicting the performance of deep learning model for each ConvLSTM variants on the different datasets.

TABLE 3 Comparison of the different state-of-the-algorithms with the proposed deep learning architecture using SCan-ConvLSTM.

Method	Hockey Fight	Crowd Violence	Movies
MoSIFT + HIK (Bermejo Nievas et al., 2011)	90.9%	-	89.5%
Vif (Hassner et al., 2012)	82.9% $\pm$ 0.14%	81.3% $\pm$ 0.21%	-
Deniz (Deniz et al., 2014)	90.1 $\pm$ 0%	-	98.0% $\pm$ 0.22%
Substantial derivative (Mohammadi et al., 2015)	-	85.43 $\pm$ 0.21%	96.89% $\pm$ 0.21%
Bilinski and Bremond (2016)	93.4%	96.4%	99%
2D CNN + Hough Forests (Serrano et al., 2018)	94.6% $\pm$ 0.6%	-	99% $\pm$ 0.5%
Rethinking 3DCNN (Li et al., 2019)	98.3 $\pm$ 0.81%	97.17 $\pm$ 0.95%	100% $\pm$ 0%
ConvLSTM (Sudhakaran & Lanz, 2017)	97.1% $\pm$ 0.55%	94.57% $\pm$ 2.34%	100% $\pm$ 0%
Bi-ConvLSTM (Hanson et al., 2019)	98.1% $\pm$ 0.58%	96.32% $\pm$ 1.52%	100% $\pm$ 0%
C3D (Tran et al., 2015)	96.5%	84.4%	100%
I3D (Flow only) (Carreira & Zisserman, 2017)	84%	88.89%	100%
I3D (Fusion) (Carreira & Zisserman, 2017)	97.50%	88.89%	100%
I3D (RGB only) (Carreira & Zisserman, 2017)	98.5%	86.67%	100%
Flow Gated Network (Cheng et al., 2019)	98.00%	88.87%	100%
SPIL (Su et al., 2020)	96.8%	94.5%	98.5%
Ours (SCan-ConvLSTM)	99.83% $\pm$ 0.52%	98.753% $\pm$ 0.63%	100% $\pm$ 0%

TABLE 4 Comparison of the state-of-the-art algorithms on RWF-2000 dataset.

Method	Accuracy (%)
ConvLSTM (Sudhakaran & Lanz, 2017)	77
C3D (Tran et al., 2015)	82.75
I3D (Flow only) (Carreira & Zisserman, 2017)	75.50
I3D (RGB only) (Carreira & Zisserman, 2017)	85.75
I3D (TwoStream) (Carreira & Zisserman, 2017)	81.50
Flow Gated Network (Cheng et al., 2019)	87.25
SPIL-space (Su et al., 2020)	86.4
SPIL-mask (Su et al., 2020)	89.3
Ours (SCan-ConvLSTM)	85.97

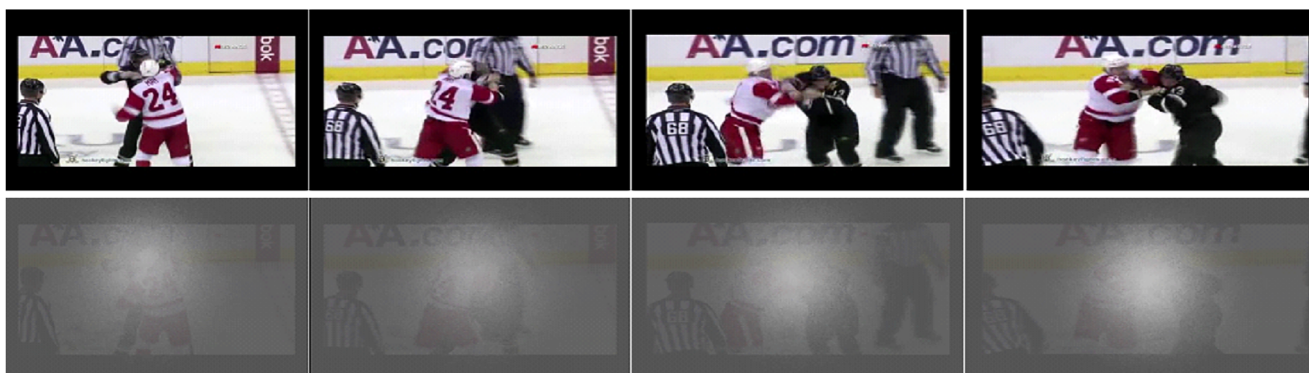
In Figure 6, the graphical depiction of obtained performances of the proposed work model variations—Scan-ConvLSTM, SAtt-ConvLSTM, ConvLSTM, and C3D (Cheng et al., 2019), SPIL (Su et al., 2020) are presented, which demonstrate superior performance.

5 | CONCLUSIONS

In this work, we have successfully proposed a novel deep learning model based on Spatiotemporal architecture. The proposed deep learning architecture enhances the ability to draw better decisions by laying emphasis on the specific regions and suppressing irrelevant information among different channels from a video sequence. The performance is validated on the four different benchmark datasets. According to the experimental results, our framework was able to outperform the existing state-of-the-art algorithms on the Hockey Fight, Crowd Violence, and Movies datasets and performs reasonably well on the RWF-2000 dataset, which is more complex than the aforementioned datasets. The proposed study enables us to further improve specifically upon the recent use of attention in deep learning to fight detection in videos. In the future, one can extend the scope of the proposed system to the other violent activities apart from human violence and investigate further improvements for better efficiency and robustness.



(a)



(b)

FIGURE 5 Qualitative results on SCan-ConvLSTM embedded with the spatio-temporal architecture using (a) RWF-2000 and (b) Hockey Fight: The first and third rows represents frames from the aforementioned datasets, and second and fourth rows show the saliency predictions. Here, white region indicates the area of focus, that is, the fight scene.

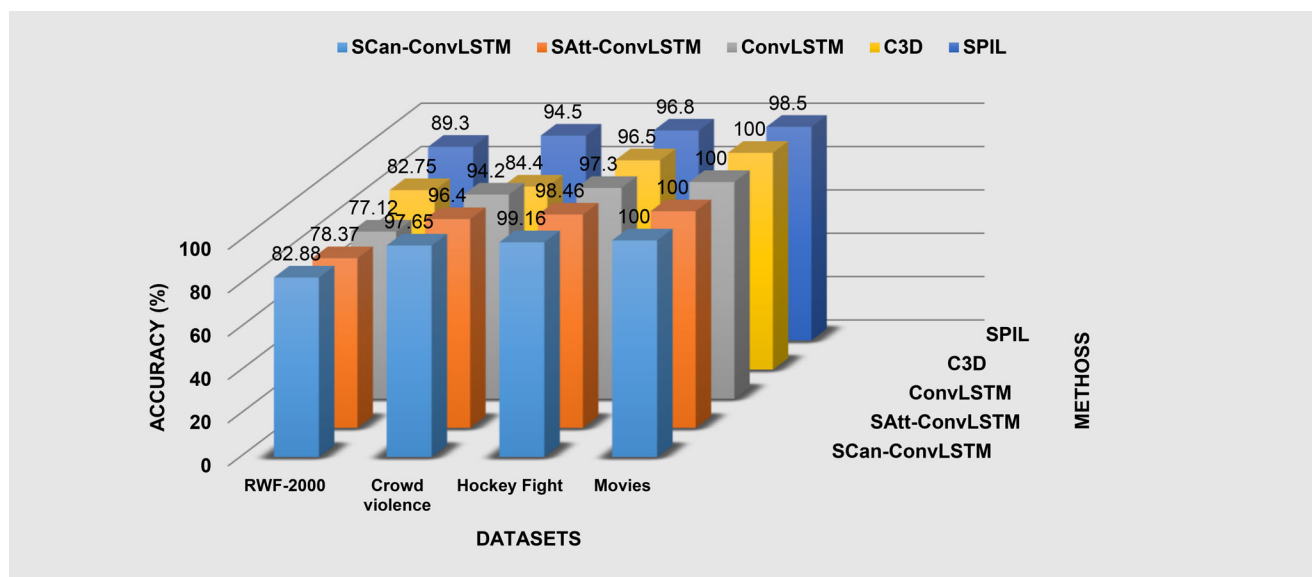


FIGURE 6 The graphical depiction of obtained performance of the proposed work and other-state-of-the-arts.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

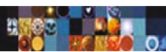
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REFERENCES

- Amin, J., Anjum, M. A., Ibrar, K., Sharif, M., Kadry, S., & Crespo, R. G. (2023). Detection of anomaly in surveillance videos using quantum convolutional neural networks. *Image and Vision Computing*, 135, 104710. <https://doi.org/10.1016/j.imavis.2023.104710>
- Bermejo Nievas, E., Deniz Suarez, O., Bueno García, G., & Sukthankar, R. (2011). Violence detection in video using computer vision techniques. *Lecture Notes in Computer Science*, 6855(PART 2), 332–339. https://doi.org/10.1007/978-3-642-23678-5_39
- Bilinski, P., & Bremond, F. (2016). Human violence recognition and detection in surveillance videos. In *2016 13th IEEE international conference on advanced video and signal based surveillance (AVSS)* (pp. 30–36). IEEE. <https://doi.org/10.1109/AVSS.2016.7738019>
- Carreira, J., & Zisserman, A. (2017). Quo vadis, action recognition? A new model and the kinetics dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 6299–6308). IEEE.
- Chang, Y., Tu, Z., Xie, W., & Yuan, J. (2020). *Clustering driven deep autoencoder for video anomaly detection* (pp. 329–345). Springer.
- Chen, L., Zhang, H., Xiao, J., Nie, L., Shao, J., Liu, W., & Chua, T.-S. (2016). SCA-CNN: Spatial and channel-wise attention in convolutional networks for image captioning. IEEE.
- Cheng, M., Cai, K., & Li, M. (2019). RWF-2000: An open large scale video database for violence detection. IEEE.
- Cheng, M., Cai, K., & Li, M. (2021). RWF-2000: An open large scale video database for violence detection. In *2020 25th International Conference on Pattern Recognition (ICPR)* (pp. 4183–4190). ICPR. <https://doi.org/10.1109/ICPR48806.2021.9412502>
- Coletto, M., Lucchese, C., & Orlando, S. (2018). Do violent people smile (pp. 1465–1468). ACM. <https://doi.org/10.1145/3184558.3191594>
- Deniz, O., Serrano, I., Bueno, G., & Kim, T. K. (2014). Fast violence detection in video. In *VISAPP 2014—Proc. 9th int. conf. comput. vis. theory appl.* (Vol. 2, pp. 478–485). VISAPP. <https://doi.org/10.5220/0004695104780485>
- Ding, C., Fan, S., Zhu, M., Feng, W., & Jia, B. (2014). Violence detection in video by using 3D convolutional neural networks (pp. 551–558). Springer.
- Gracia, I. S., Suarez, O. D., Garcia, G. B., & Kim, T. K. (2015). Fast fight detection. *PLoS One*, 10(4), 1–19. <https://doi.org/10.1371/journal.pone.0120448>
- Hanson, A., Pnvr, K., Krishnagopal, S., & Davis, L. (2019). Bidirectional convolutional LSTM for the detection of violence in videos (pp. 280–295). Springer.
- Hassner, T., Itcher, Y., & Kliper-Gross, O. (2012). Violent flows: Real-time detection of violent crowd behavior. In *2012 IEEE computer society conference on computer vision and pattern recognition workshops* (pp. 1–6). IEEE. <https://doi.org/10.1109/CVPRW.2012.6239348>
- He, K., Zhang, X., Ren, S., & Sun, J. (2015). Deep residual learning for image recognition. IEEE.
- Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780. <https://doi.org/10.1162/neco.1997.9.8.1735>
- Hu, J., Shen, L., Albanie, S., Sun, G., & Wu, E. (2017). Squeeze-and-excitation networks. IEEE.
- Irfanullah, Hussain, T., Iqbal, A., Yang, B., & Hussain, A. (2022). Real time violence detection in surveillance videos using convolutional neural networks. *Multimedia Tools & Applications*, 81, 38151–38173. <https://doi.org/10.1007/s11042-022-13169-4>
- Khaire, P., & Kumar, P. (2022). A semi-supervised deep learning based video anomaly detection framework using RGB-D for surveillance of real-world critical environments. *Forensic Science International: Digital Investigation*, 40, 301346. <https://doi.org/10.1016/j.fsidi.2022.301346>
- Kim, J., & Canny, J. (2017). Interpretable learning for self-driving cars by visualizing causal attention. In *Proceedings of the IEEE international conference on computer vision* (Vol. 2017, pp. 2961–2969). IEEE. <https://doi.org/10.1109/ICCV.2017.320>
- Le, V. T., & Kim, Y. G. (2023). Attention-based residual autoencoder for video anomaly detection. *Applied Intelligence*, 53, 3240–3254. <https://doi.org/10.1007/s10489-022-03613-1>
- Li, J., Jiang, X., Sun, T., & Xu, K. (2019). Efficient violence detection using 3D convolutional neural networks. In *2019 16th IEEE international conference on advanced video and signal based surveillance (AVSS)* (pp. 1–8). IEEE. <https://doi.org/10.1109/AVSS.2019.8909883>
- Li, Q., Yang, R., Xiao, F., Bhanu, B., & Zhang, F. (2022). Attention-based anomaly detection in multi-view surveillance videos. *Knowledge-Based Systems*, 252, 109348. <https://doi.org/10.1016/j.knosys.2022.109348>
- Li, Z., Gavriluyk, K., Gavves, E., Jain, M., & Snoek, C. G. M. (2018). VideoLSTM convolves, attends and flows for action recognition. *Computer Vision and Image Understanding*, 166, 41–50. <https://doi.org/10.1016/j.cviu.2017.10.011>
- Mohammadi, S., Kiani, H., Perina, A., & Murino, V. (2015). Violence detection in crowded scenes using substantial derivative. In *2015 12th IEEE international conference on advanced video and signal based surveillance (AVSS)* (pp. 1–6). IEEE. <https://doi.org/10.1109/AVSS.2015.7301787>
- Peixoto, B., Lavi, B., Bestagini, P., Dias, Z., & Rocha, A. (2020). Multimodal violence detection in videos. In *ICASSP 2020–2020 IEEE international conference on acoustics, speech and signal processing (ICASSP)* (pp. 2957–2961). ICASSP. <https://doi.org/10.1109/ICASSP40776.2020.9054018>
- Serrano, I., Deniz, O., Espinosa-Aranda, J. L., & Bueno, G. (2018). Fight recognition in video using Hough forests and 2D convolutional neural network. *IEEE Transactions on Image Processing*, 27(10), 4787–4797. <https://doi.org/10.1109/TIP.2018.2845742>
- Sharma, S., Kiros, R., & Salakhutdinov, R. (2015). Action recognition using visual attention (pp. 1–11). IEEE.
- Su, Y., Lin, G., Zhu, J., & Wu, Q. (2020). Human interaction learning on 3D skeleton point clouds for video violence recognition (pp. 74–90). Springer.
- Sudhakaran, S., & Lanz, O. (2017). Learning to detect violent videos using convolutional long short-term memory. In *2017 14th IEEE international conference on advanced video and signal based surveillance (AVSS)* (pp. 1–6). IEEE. <https://doi.org/10.1109/AVSS.2017.8078468>
- Tran, D., Bourdev, L., Fergus, R., Torresani, L., & Paluri, M. (2015). Learning spatiotemporal features with 3d convolutional networks. In *Proceedings of the IEEE international conference on computer vision* (pp. 4489–4497). IEEE.
- Ullah, F. U. M., Ullah, A., Muhammad, K., Haq, I. U., & Baik, S. W. (2019). Violence detection using spatiotemporal features with 3D convolutional neural network. *Sensors*, 19(11), 2472. <https://doi.org/10.3390/s19112472>
- Ullah, W., Ullah, F. U. M., Khan, Z. A., & Baik, S. W. (2023). Sequential attention mechanism for weakly supervised video anomaly detection. *Expert Systems with Applications*, 230, 120599. <https://doi.org/10.1016/j.eswa.2023.120599>
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukjin, I. (2017). Attention is all you need. In *Advances in neural information processing systems* (Vol. 2017, pp. 5999–6009). MIT Press.
- Yan, C., Hao, Y., Li, L., Yin, J., Liu, A., Mao, Z., Chen, Z., & Gao, X. (2021). Task-adaptive attention for image captioning. *IEEE Transactions on Circuits and Systems for Video Technology*, 32, 1–51. <https://doi.org/10.1109/TCSVT.2021.3067449>
- Yan, C., Tu, Y., Wang, X., Zhang, Y., Hao, X., Zhang, Y., & Dai, Q. (2020). Corrections to “STAT: Spatial-temporal attention mechanism for video captioning”. *IEEE Transactions on Multimedia*, 22(3), 830. <https://doi.org/10.1109/TMM.2020.2966830>



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